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Order Book Basics Primer

Introduction

An exchange is a marketplace that matches buyers and sellers with each other. Market participants can send "orders" to an exchange to indicate their interest in trading a security at a specific quantity and price. Participants who want to buy can send a "bid" (an order to buy) and participants who want to sell can send an "offer" (an order to sell).

So for example, a bid for 100 shares of ABC at $10.00 means the buyer is willing to buy 100 shares of ABC stock at $10.00/share (or lower), and an offer for 100 shares of ABC at $10.00 means the seller is willing to sell 100 shares of ABC at $10.00/share (or higher).

A bid and an offer whose prices are the same or "cross" will pair, resulting in a "trade" (also called an "execution"). We consider bid and offer prices to "cross" when the bid price is higher than the offer price; in those cases, a trade will occur at the price of the order that was already listed on the exchange (the "resting" order). Unpaired bids and offers make up the "order book," which is maintained by the exchange.

When there are multiple bids and offers that can be matched with each other, the exchange needs to determine the priority of which bids and offers to match first. Assume this is done using "price-time priority". In price-time priority, the exchange will prioritize orders with more competitive prices. Higher bid prices are more competitive than lower ones, and lower offer prices are more competitive than higher ones. If multiple orders have equally competitive prices, the exchange will prioritize whichever of those orders was placed earliest. These two factors determine who is the most willing buyer (for bid) or seller (for offer) and therefore should be matched first.

Please see the next page for some worked examples.

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Worked Example - Basics

Suppose you sent a bid to the exchange of $15.00 for 100 ABC shares. Also suppose that Anne (another market participant) sent an offer of 100 shares at $16.00.

The order book for stock ABC would be as follows:

Bid Quantity	Bid Price	Ask Price	Ask Quantity
100	$15.00	$16.00	100

There is no trade because the bid and offer prices do not equal or cross. Now, suppose that Bob (another market participant) sends an offer of 100 shares at $15.00.

Bid Quantity	Bid Price	Ask Price	Ask Quantity
100	$15.00	$15.00	100
		$16.00	100

Because Bob's offer price is less than or equal to your bid price, a trade will occur: you buy 100 shares from Bob at a price of $15.00. While we're showing the offer on the order book here, in reality, the trade occurs instantaneously, and the offer never truly shows up on the order book. Instead, it is immediately paired with the bid and the bid falls off the book.

After this trade, your bid and Bob's offer will no longer appear on the order book because both orders have been paired. Anne's offer of 100 shares at $16.00 will remain on the order book because it is unpaired. If another participant later sends a bid with a price equal to or greater than Anne's offer price, another trade would occur.

Worked Example - Price Priority

Consider the below order book for ABC:

Bid Quantity	Bid Price	Ask Price	Ask Quantity
100	$15.00	$16.00	100
100	$14.00	$17.00	100

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If Anne sends a bid for 100 shares at $17.00 then we have two plausible offers against which her bid could execute, the offer at $16.00 and the offer at $17.00. To break this tie, we apply the concept of “price-time priority” and note that the $16.00 offer is a better price and therefore we execute against that offer for the full quantity of 100 shares. We execute at a price of $16.00, as this corresponds to the price of the “resting order”. We do not need to consider time priority, as price priority alone was sufficient to break our tie.

Worked Example - Multiple Executions and Partial Executions

Consider the below order book for ABC:

Bid Quantity	Bid Price	Ask Price	Ask Quantity
100	$15.00	$16.00	100
50	$14.00	$17.00	50

If Bob sends an offer of 200 shares at $14.00 then we have two plausible bids against which his offer could be executed, the bid at $15.00 and the bid at $14.00. In this case, we note that the quantity of the offer is such that we will execute against both of the resting bids. This will result in executions of 100 shares at $15.00 and 50 shares at $14.00, executing in both cases at the price of the resting order.

Bob’s offer is not completely fulfilled though, as he has only executed 150 shares, but his initial offer was for 200 shares. This is known as a “partial execution”, and the remaining 50 shares of Bob’s offer will be placed into the order book. The state of the order book after Bob’s offer and subsequent executions is shown below:

Bid Quantity	Bid Price	Ask Price	Ask Quantity
		$14.00	50
		$16.00	100
		$17.00	50

Worked Example - Time Priority

Consider the below order book for ABC. We note that we have two separate offers at $16.00. The topmost offer was placed by Anne at 1.30PM and the bottommost was placed by Bob at 2.30PM.*

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Bid Quantity	Bid Price	Ask Price	Ask Quantity
100	$15.00	$16.00	150
100	$14.00	$16.00	200

If you sent an order to buy 100 shares at $16.00, then we have two plausible offers which your bid could execute against, both of which are at $16.00. To break this tie we apply price-time priority. We note that both offers have the same price, so we must examine the time of the offers to determine which offer we should match against. Anne's order is the oldest, which means we give her order priority and therefore we execute 100 shares at $16.00 against Anne's offer.

Summary

We have seen that sending a single order may result in a variety of outcomes, including:

- No executions
- A single execution
- Multiple executions
- Partial executions

Additionally, we have seen that the pricing rules used by an exchange may result in the execution price being different to the price of the offer sent into the order book.

We have also seen the application of "price-time priority", and how that dictates which offers will be paired and executed.